

```
import numpy as np
import pandas as pd
```

```
def sigmoid(x):
    return 1 / (1 + np.exp(-x))

def sigmoid_derivative(x):
    return x * (1 - x)
```

```
X = np.array([[0,0,1],
              [0,1,1],
              [1,0,1],
              [1,1,1]])

y = np.array([[0],
              [1],
              [1],
              [0]])
```

```
np.random.seed(1)
w0 = 2 * np.random.random((3,4)) - 1
w1 = 2 * np.random.random((4,1)) - 1
```

```
errors = []
for i in range(5000):

    a0 = X
    a1 = sigmoid(np.dot(a0,w0))
    a2 = sigmoid(np.dot(a1,w1))

    output_error = a2 - y

    errors.append({
        "epochs": i,
        "error": np.mean(np.abs(output_error))
    })
    if (i % 500) == 0:
        print("Error after "+str(i)+" epochs:" + str(np.mean(np.abs(output_error))))
        layer2_delta = output_error * sigmoid_derivative(a2)
        layer1_error = layer2_delta.dot(w1.T)
        layer1_delta = layer1_error * sigmoid_derivative(a1)

        w1 -= a1.T.dot(layer2_delta)
        w0 -= a0.T.dot(layer1_delta)
```

```
Error after 0 epochs:0.4964100319027255
Error after 500 epochs:0.099292092737883
Error after 1000 epochs:0.042984647810726195
Error after 1500 epochs:0.030122764096817193
Error after 2000 epochs:0.024121949891468507
Error after 2500 epochs:0.020542517228580394
Error after 3000 epochs:0.018121400645678867
Error after 3500 epochs:0.01635335860503167
Error after 4000 epochs:0.014993830283221644
Error after 4500 epochs:0.013908868267333547
```

```
pd.DataFrame(errors).plot(x='epochs', y='error', title = 'Error in the network while training the model')
```

<Axes: title={'center': 'Error in the network while training the model'}, xlabel='epochs'>

